Pre-requisites

ComicCraft - AI Comic Story Creator using Gemini Models

1. Overview

Before building ComicCraft, you need a working knowledge of the tools and libraries the project is built on. ComicCraft is a FastAPI web application that uses Google Gemini models for story text, Stable Diffusion (through Hugging Face Diffusers) for illustrations, Jinja2 templates for the web pages, and FPDF for exporting the finished comic as a PDF. The project document lists ten pre-requisites, explained one by one below.

2. The 10 pre-requisites and why each one matters

#	Pre-requisite	Role in ComicCraft	Reference
1	FastAPI Framework Knowledge	Backend framework. Handles routing, form input, JSON input and template rendering (routes.py).	devdocs.io/fastapi
2	Uvicorn ASGI Server	Runs the FastAPI app locally with hot reload: `uvicorn app.main:app --reload`	uvicorn.org
3	Python Programming Proficiency	All logic (routes, AI calls, layout builder, exporter) is written in Python.	docs.python.org/3
4	HTML, CSS and Jinja2 Template Skills	Builds index.html, comic_preview.html and export_success.html and binds backend data into them.	w3schools.com
5	Google Gemini API Familiarity	Gemini Flash creates the 5-panel outline; Gemini Pro writes narration and dialogue.	ai.google.dev
6	Hugging Face Diffusers and Stable Diffusion	Generates comic-style images from each panel's image prompt.	huggingface.co/docs/diffusers
7	PyTorch Basics	Deep-learning backend that Stable Diffusion runs on.	pytorch.org/docs/stable
8	PDF Generation with FPDF	save_pdf() in exporters.py puts panel images and narration on PDF pages.	fpdf.org
9	Version Control with Git	Track changes to the project code and collaborate safely.	git-scm.com/docs/git
10	Python Package and Environment Setup (pip, virtualenv or conda)	Keeps project libraries isolated in a virtual environment.	virtualenv.pypa.io

3. What to know in each area

FastAPI and Uvicorn

- How to declare a route (GET and POST) and attach a handler function.
- How to read HTML form fields using `Form(...)` parameters.
- How to validate JSON request bodies using a Pydantic model (ComicCraft uses `PromptRequest`).

- How to return an HTML page from a template or a JSON response, and how to raise `HTTPException` on errors.
- FastAPI gives free interactive API docs at `/docs`, useful for testing.

Python

- Functions, dictionaries and lists (panel outlines are a list of dictionaries).
- Modules and imports: the project is split into gemini_flash.py, gemini_pro.py, image_generator.py, layout_builder.py, exporters.py and routes.py.
- try/except for handling API and JSON errors, file paths and saving files.

HTML, CSS and Jinja2

- Semantic HTML forms with labelled inputs (story prompt, character name, setting, tone, art style).
- CSS for a clean, responsive layout with a scenic background.
- Jinja2 loops and variables to display each panel's title, image, description, caption and narration.

Gemini API

- Getting and using an API key (kept in an environment variable, never hard-coded).
- Difference between the two models used: **gemini-1.5-flash** (fast, structured output) and **gemini-1.5-pro** (detailed, creative text).
- Writing prompts that ask for structured output (panel number, title, scene description, image prompt).

Stable Diffusion, Diffusers and PyTorch

- Model used: **runwayml/stable-diffusion-v1-5**.
- How a text prompt becomes an image, and how the image is saved into `static/panels`.
- Basic PyTorch ideas (tensors, running a model on CPU or GPU) so you can troubleshoot speed and memory.

FPDF, Git, virtual environments

- FPDF: adding pages, placing images and writing multi-line text; final file saved to `static/exports` with a timestamped name.
- Git: commit, branch and push so your work is safe.
- venv: create, activate and install packages inside the environment.

4. Environment setup steps (from Activity 1.3)

These commands prepare your machine to meet the pre-requisites practically:

Step 1 - Make sure Python and pip are installed

Check with `python --version` and `pip --version`.

Step 2 - Create and activate a virtual environment

```
python -m venv comiccraft-env
comiccraft-env\Scripts\activate     # Windows
```

Step 3 - Install the required libraries

```
pip install fastapi uvicorn jinja2 python-multipart google-generativeai \
    diffusers transformers fpdf Pillow accelerate
```

Step 4 - Run the server

```
uvicorn app.main:app --reload
```

Open `http://127.0.0.1:8000` for the app and `/docs` for API testing.

Step 5 - Create the project structure

Set up the initial project folders for ComicCraft (app code, templates, static/panels for images and static/exports for PDFs).

5. Quick self-check list

- I can create and activate a virtual environment.
- I can install packages with pip without errors.
- I understand how a FastAPI route receives form data and JSON.
- I have a Gemini API key and a Hugging Face token ready.
- I know the difference between Gemini Flash and Gemini Pro in this project.
- I can render a simple Jinja2 template from FastAPI.
- I can create a basic PDF using FPDF.
- Git is installed and I can make a commit.

> **Tip:** If you are new to any item, start with its documentation link in the table above and do one small practice example before beginning Milestone 1.